

ITA0608 - Machine Learning

List of Lab Exercises

5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

same using appropriate data sets.

Aim

To implement the **K-Nearest Neighbours (K-NN)** algorithm using the Iris dataset and classify test samples.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Create a K-NN classifier ($k = 3$).
5. Train the classifier using the training data.
6. Predict the class labels for the test data.
7. Calculate the accuracy.
8. Display the confusion matrix.

Program Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
```

```

X, y, test_size=0.3, random_state=1

)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Values:", y_pred)

print("Actual Values   :", y_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy =", accuracy * 100, "%")

new_sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_sample)

print("Prediction for New Sample:", iris.target_names[prediction[0]])

```

Output

```

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Values:", y_pred)
print("Actual Values   :", y_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy =", accuracy * 100, "%")

new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)

print("Prediction for New Sample:", iris.target_names[prediction[0]])

```

```

... Predicted Values: [0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 1 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 1 2 1]
Actual Values   : [0 1 1 0 2 1 2 0 0 2 1 0 2 1 1 0 1 1 0 0 1 1 1 0 2 1 0 0 1 2 1 2 1 2 2 0 1
 0 1 2 2 0 2 2 1]
Accuracy = 97.77777777777777 %
Prediction for New Sample: setosa

```

Result

The K-Nearest Neighbours (K-NN) classifier was implemented successfully. The model classified the Iris flowers with high accuracy, and the confusion matrix showed the prediction performance.